

Analysis of Machining Parameters on Surface Roughness in CNC Turning Processes

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1. Introduction and Data Source

The dataset selected for this project is “**CNC turning: roughness, forces and tool wear**”, obtained from Kaggle. This dataset was developed from controlled machining experiments conducted at the Competence Center in Manufacturing (CCM) at the Aeronautics Institute of Technology (ITA). The purpose of the experiments was to study how machining parameters influence key performance measures such as surface roughness, cutting forces, and tool wear.

Because the data were collected under controlled laboratory conditions where factors were deliberately set by the experimenters, this dataset represents an **experimental study** rather than an observational one. This makes it especially appropriate for analysis using ANOVA techniques, as causal relationships between factors and the response variable can be investigated.

2. Description of Variables

Response Variable

- **Surface Roughness (Ra):** A continuous numerical variable measuring the average deviation of the machined surface (in micrometers). Lower values indicate smoother surfaces.
-

Factors (Independent Variables)

- **Depth of Cut (ap):** 0.25, 0.5, 0.8 mm
 - **Cutting Speed (vc):** 310, 350, 390 m/min
 - **Feed Rate (f):** 0.07, 0.10, 0.13 mm/rev
-

Additional Variables

- **Replica:** Indicates repeated experimental runs for the same factor combination
 - **Position:** Measurement location on the machined surface
 - **Run:** Constructed variable representing each unique experimental run
-

3. Data Structure and Number of Observations

The experimental design is a full factorial design:

$$3 \times 3 \times 3 = 27 \text{ treatment combinations}$$

Each combination has:

- 2 replicates
- 6 measurements per replicate

Total number of observations:

$$27 \times 2 \times 6 = 324$$

The design is **balanced at the treatment level**, meaning each factor combination has the same number of observations.

4. Potential Issues in the Data

- **Non-independence:** Multiple measurements taken within the same replicate (run) are likely correlated, since they are collected under identical machining conditions. Treating these observations as independent would violate model assumptions.
- **Hierarchical structure:** The data exhibit a nested structure, where measurements are nested within replicates, and replicates are nested within treatment combinations ($ap \times vc \times f$). This induces dependence that must be accounted for in the modeling process.
- **Heteroscedasticity:** Preliminary diagnostics indicate that the variability of surface roughness is not constant across factor levels, particularly with respect to feed rate. This violates the assumption of constant variance required by standard ANOVA models.

Ignoring these features would lead to underestimated variability, biased standard errors, and invalid statistical inferences .

5. Sample of the Data

ap	vc	f	Ra	Replica	Position
0.25	310	0.07	0.85	1	1
0.25	310	0.07	0.88	1	2

ap	vc	f	Ra	Replica	Position
0.25	310	0.07	0.90	1	3
0.25	310	0.07	0.87	1	4
0.25	310	0.07	0.89	1	5
0.25	310	0.07	0.86	1	6

(This is a subset of one treatment combination; full dataset contains 324 observations.)

6. Proposed Analysis

The analysis will use a **mixed-effects factorial ANOVA model**.

- Fixed effects: ap, vc, f
- Random effect: Run (to account for repeated measurements)

Model:

$$y_{ijkl} = \mu + A_i + B_j + C_k + (AB)_{ij} + (AC)_{ik} + (BC)_{jk} + (ABC)_{ijk} + R_{l(ijk)} + \epsilon_{ijkl}$$

This model allows for:

- Testing main effects
 - Evaluating interaction effects
 - Accounting for within-run variability
-

7. Research Questions and Hypotheses

1. Does depth of cut affect surface roughness?
2. Does cutting speed affect surface roughness?
3. Does feed rate affect surface roughness?
4. Are there interaction effects between factors?
5. Which factor has the strongest influence?

Each question corresponds to a hypothesis test where:

- H_0 : No effect
 - H_a : Significant effect
-

8. Summary

This dataset provides a well-structured experimental framework for analyzing machining processes. Its factorial design combined with repeated measurements makes it ideal for mixed-effects ANOVA. The results will provide insight into how machining parameters influence surface roughness and demonstrate proper statistical modeling of hierarchical data.

9. R Code: Data Preparation, EDA, and Modeling

```
In [ ]: library(dplyr)

# Load data
data <- read.csv("/kaggle/input/datasets/adorigueto/cnc-turning-roughness-forces-and-tool-wear/E)

# Convert to factors
data <- data %>%
  mutate(across(c(ap, vc, f, Replica, Position), as.factor))

# Keep relevant variables
data <- data %>%
  select(ap, vc, f, Ra, Replica, Position)

# Create Run ID
data$Run <- interaction(data$ap, data$vc, data$f, data$Replica)
```

We load the dataset, select relevant variables, convert key machining parameters into factors, and create a unique Run identifier to represent each experimental combination for use in mixed-effects modeling.

```
In [ ]: # Check missing values
colSums(is.na(data))
```

We first check for missing values in the dataset by counting the number of NA entries in each column. It helps identify data quality issues before model fitting and ensures that no variables contain missing observations that could affect the analysis. In this dataset, there are no missing values.

```
In [19]: library(ggplot2)
library(patchwork)

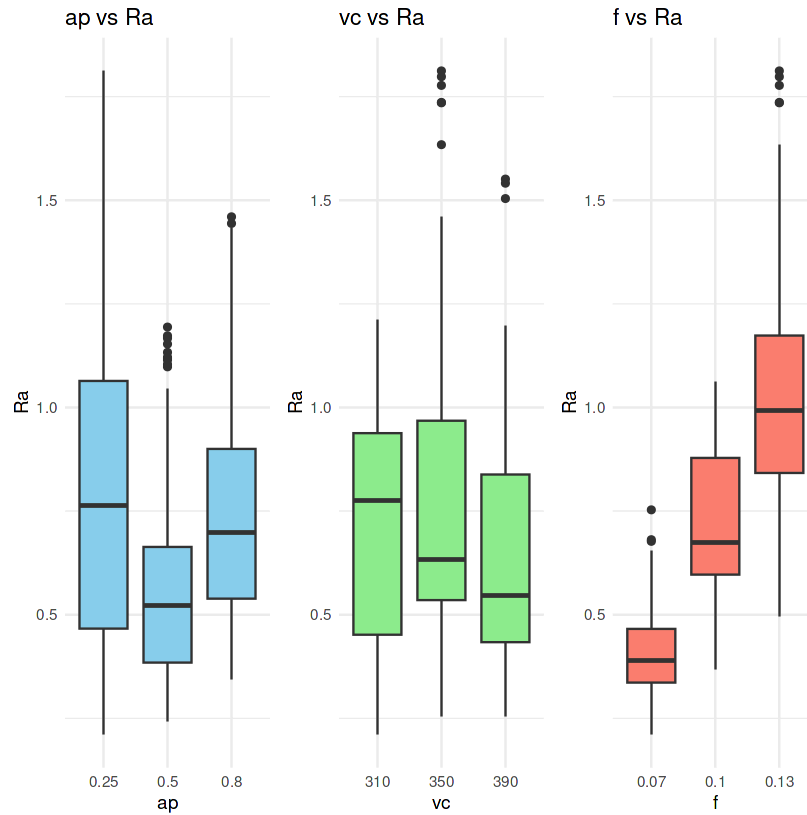
theme_set(theme_minimal())

p1 <- ggplot(data, aes(x = ap, y = Ra)) +
  geom_boxplot(fill = "skyblue") +
  labs(title = "ap vs Ra", x = "ap", y = "Ra")

p2 <- ggplot(data, aes(x = vc, y = Ra)) +
  geom_boxplot(fill = "lightgreen") +
  labs(title = "vc vs Ra", x = "vc", y = "Ra")

p3 <- ggplot(data, aes(x = f, y = Ra)) +
  geom_boxplot(fill = "salmon") +
  labs(title = "f vs Ra", x = "f", y = "Ra")
```

```
(p1 | p2 | p3) +  
plot_layout(widths = c(1, 1, 1))
```



Above are boxplots of surface roughness (Ra) against each machining parameter (ap, vc, and f). A minimal theme is applied for clean visualization, and the combined figure allows easy comparison of how each factor influences Ra.

```
In [65]: library(lme4)  
  
# Fit mixed-effects model  
model <- lmer(Ra ~ ap * vc * f + (1 | Run), data = data)  
  
# ANOVA table  
anova(model)
```

A anova: 7 × 4				
	npars	Sum Sq	Mean Sq	F value
	<int>	<dbl>	<dbl>	<dbl>
ap	2	0.060188834	0.030094417	6.6695746
vc	2	0.016717041	0.008358521	1.8524292
f	2	0.426138058	0.213069029	47.2207114
ap:vc	4	0.037956573	0.009489143	2.1029996
ap:f	4	0.022980380	0.005745095	1.2732375
vc:f	4	0.006694608	0.001673652	0.3709175
ap:vc:f	8	0.023094621	0.002886828	0.6397835

This analysis fits a mixed-effects model to explain surface roughness (Ra) using machining parameters and their interactions, while accounting for variability between experimental runs.

10. Model Effects

10.1 Main Effects

Depth of Cut (ap)

- F = 6.67, p = 0.0044

Depth of cut has a statistically significant effect on surface roughness. This indicates that variations in cutting depth produce systematic changes in surface quality even after accounting for within-run variability.

Cutting Speed (vc)

- F = 1.85, p = 0.176

Cutting speed is not statistically significant. Its effect on surface roughness is not distinguishable from experimental variability within the current design.

Feed Rate (f)

- F = 47.22, p ≈ 1.53 × 10⁻⁹

Feed rate is the dominant factor in the model. The extremely strong statistical significance indicates a robust and highly consistent effect on surface roughness across all experimental conditions.

10.2 Interaction Effects

ap × vc

- $F = 2.10, p = 0.108$

ap × f

- $F = 1.27, p = 0.305$

vc × f

- $F = 0.37, p = 0.827$

ap × vc × f

- $F = 0.64, p = 0.738$

None of the interaction terms are statistically significant.

Interpretation

The absence of significant interaction effects suggests that the machining parameters operate largely in an **additive manner** within the experimental region. That is, the effect of each factor on surface roughness does not meaningfully depend on the levels of the other factors.

11. Model Assumptions and Diagnostic Checks

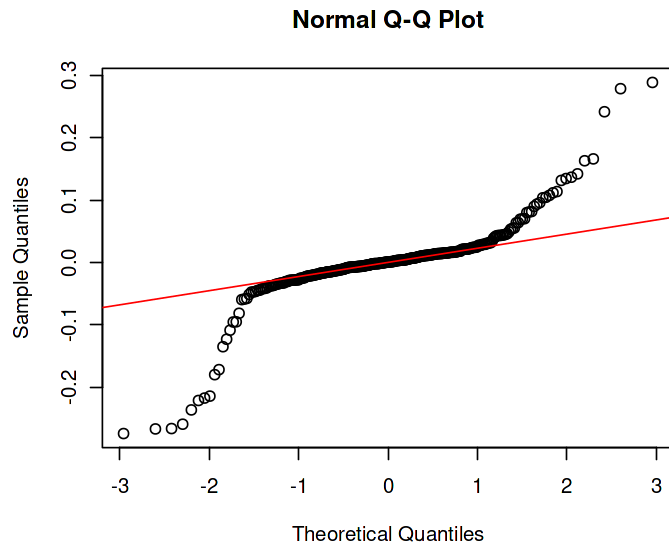
Before final interpretation of the mixed-factorial model results, the validity of model assumptions is evaluated. These checks ensure that statistical inference based on the model is appropriate and reliable.

11.1 R Code: Preparation for Diagnostic Checks

```
In [ ]: # Extract residuals and fitted values from mixed model
residuals_m <- resid(model)
fitted_m <- fitted(model)
```

11.2 Assumption: Normality of Residuals

```
In [59]: # 2. Normality check (Q-Q plot)
qqnorm(residuals_m, main = "Normal Q-Q Plot")
qqline(residuals_m, col = "red")
```



The Q-Q plot of residuals shows that the central portion of the data aligns closely with the theoretical normal distribution line, indicating that the bulk of the residuals are approximately normally distributed.

However, some deviations are observed in the tails, where points slightly diverge from the reference line. This pattern suggests mild tail non-normality, which is common in experimental measurement data due to occasional extreme observations or process variability.

Importantly, because the central region — where most observations lie — adheres closely to the diagonal reference line, the normality assumption is considered reasonably satisfied for the purposes of inference in the mixed-factorial model.

Given the robustness of mixed-effects models to mild deviations from normality, especially in balanced experimental designs, these minor tail departures are not expected to materially affect inference.

11.2 Assumption: Constant Variance (Homoscedasticity)

```
In [ ]: # Load package for Levene's test
library(car)

# Levene's test for homogeneity of variance across main factor combinations

leveneTest(Ra ~ ap * vc * f, data = data)
```

Levene's test for homogeneity of variance across treatment combinations is highly significant ($F(26, 297) = 13.86, p < 0.001$).

This provides strong statistical evidence that the variance of surface roughness is not constant across the different combinations of machining parameters (ap, vc, f).

11.4 Assumption: Independence via Random Effects

Independence is addressed through the inclusion of a random intercept for experimental runs in the mixed-effects model.

This structure accounts for correlation among repeated measurements taken under identical machining conditions, ensuring that residual dependence does not bias fixed-effect estimates or standard errors.

11.5 Summary of Diagnostic Checks

The diagnostic analysis indicates that several key assumptions of the mixed-factorial model are only partially satisfied.

Residual diagnostics suggest that normality is reasonably adequate, with only mild deviations in the tails. However, more serious concerns arise regarding variance homogeneity.

12. Relationship Between Machining Parameters and Variance of Surface Roughness

To better understand the source of heteroscedasticity in the machining process, this section examines whether the variability in surface roughness depends on the full set of machining parameters: depth of cut (ap), cutting speed (vc), and feed rate (f). Rather than assuming a single driver of variance, we evaluate whether multiple factors contribute to changes in dispersion.

12.1 Methodology: Residual-Based Variance Modeling

Variance is assessed indirectly using the absolute values of normalized residuals from the fitted mixed-effects model. These values act as a proxy for local variability in surface roughness:

$$|e_i| \approx \text{local variability of Ra}$$

To identify potential sources of heteroscedasticity, regression models are fitted using each machining parameter individually and jointly.

12.2 R Code: Variance Dependence on All Machining Parameters

```
In [88]: # Load libraries silently
suppressPackageStartupMessages({
  library(nlme)
  library(modelsummary)
  library(kableExtra)
})

# Keep original data untouched
data_diag <- data.frame(data)

# Fit model
mixed_model <- lme(
  Ra ~ ap * vc * f,
```

```

random = ~1 | Run,
data = data_diag
)

# Extract normalized residuals
data_diag$res_abs <- abs(resid(mixed_model, type = "normalized"))

# Ensure feed rate is numeric (for regression trend test)
data_diag$f_num <- as.numeric(as.character(data_diag$f))

# Individual variance checks
lm_ap <- lm(res_abs ~ ap, data = data_diag)
lm_vc <- lm(res_abs ~ vc, data = data_diag)
lm_f <- lm(res_abs ~ f_num, data = data_diag)

# Combined variance model
lm_all <- lm(res_abs ~ ap + vc + f_num, data = data_diag)

# Generate the table with escape=FALSE
ms_table <- modelsummary(
  list(
    "Ap_Effect" = lm_ap,
    "Vc_Effect" = lm_vc,
    "Feed_Rate" = lm_f,
    "Combined" = lm_all
  ),
  statistic = "({std.error})",
  stars = TRUE,
  output = "kableExtra",
  escape = TRUE # Prevents double-escaping of HTML characters
) %>%
  kable_styling(bootstrap_options = c("striped", "hover"))

# HTML conversion
clean_html <- as.character(ms_table)

# Display
IRdisplay::display_html(clean_html)

```

	Ap_Effect	 Vc_Effect	Feed_Rate	Combined
(Intercept)	0.551***	0.379***	−0.229	−0.288
	(0.072)	(0.072)	(0.175)	(0.183)
ap0.5	−0.294**			−0.294**
	(0.102)			(0.096)
ap0.8	0.127			0.127
	(0.102)			(0.096)
vc350		−0.031		−0.031
		(0.102)		(0.096)
vc390		0.377***		0.377***
		(0.102)		(0.096)
f_num			7.234***	7.234***
			(1.701)	(1.608)
Num.Obs.	324	324	324	324
R2	0.053	0.058	0.053	0.164
R2 Adj.	0.047	0.053	0.050	0.151
AIC	739.1	737.2	737.0	704.5
BIC	754.2	752.3	748.3	731.0
Log.Lik.	−365.563	−364.589	−365.489	−345.252
F	8.939	9.960	18.088	12.510
RMSE	0.75	0.75	0.75	0.70

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

We compare four regression models used to explain variability in surface roughness (measured through absolute residuals): three single-factor models (ap, vc, f) and one combined model.

```
In [33]: library(effects)
library(knitr)
library(kableExtra)
library(IRdisplay)

suppressPackageStartupMessages(library(IRdisplay))

es_df <- as.data.frame(eta_squared(mixed_model, partial = TRUE))

# Generate the HTML string
html_table <- es_df %>%
  kable(format = "html", digits = 3) %>%
  kable_styling(bootstrap_options = c("striped", "hover"), full_width = FALSE) %>%
```

```
as.character()
```

```
# Force the notebook to display the HTML  
display_html(html_table)
```

Parameter	Eta2_partial	CI	CI_low	CI_high
ap	0.331	0.95	0.083	1
vc	0.121	0.95	0.000	1
f	0.778	0.95	0.636	1
ap:vc	0.238	0.95	0.000	1
ap:f	0.159	0.95	0.000	1
vc:f	0.052	0.95	0.000	1
ap:vc:f	0.159	0.95	0.000	1

The table above shows calculated partial eta-squared (η^2) effect sizes for each predictor in the model. It quantifies how much of the variability in surface roughness (Ra) is explained by each machining parameter after accounting for all other effects in the model. This helps assess the relative importance of each factor in explaining the response.

12.3 Key Results

The results from regression coefficients and partial eta-squared values describe how machining parameters influence **variability in surface roughness (Ra)**. Feed rate is the dominant source of variance, followed by depth of cut and cutting speed. The direction of influence refers to how each factor changes the level of Ra and its variability.

- **Feed rate (f):**
 - Strongest contributor to variance in Ra
 - Largest effect size ($\eta^2 = 0.778$)
 - Strong **positive effect on variance level** ($\beta = 7.234^{***}$): increasing feed rate increases surface roughness and overall variability
 - Dominant factor controlling instability in surface finish
- **Depth of cut (ap):**
 - Moderate contribution to variance ($\eta^2 = 0.331$)
 - Direction depends on level:
 - $ap = 0.5 \rightarrow$ **reduces variance in Ra** ($\beta = -0.294^{**}$)
 - $ap = 0.8 \rightarrow$ slight **increase in variance** ($\beta = 0.127$, not significant)
 - Indicates non-linear influence on process variability
- **Cutting speed (vc):**
 - Smaller contribution to variance ($\eta^2 = 0.121$)

- Directional effect varies:
 - $vc = 350 \rightarrow$ negligible or slight **reduction in variance** ($\beta = -0.031$)
 - $vc = 390 \rightarrow$ clear **increase in variance** ($\beta = 0.377^{***}$)
- Higher cutting speeds tend to increase instability in surface roughness
- **Interactions:**
 - $ap \times vc$ shows meaningful contribution to variance ($\eta^2 = 0.238$), indicating combined effects on process stability
 - $ap \times f$ and $vc \times f$ show moderate to weak contributions ($\eta^2 \approx 0.159$ and 0.052)
 - Three-way interaction contributes modestly ($\eta^2 \approx 0.159$)
- **Overall interpretation:**
 - Single-factor models explain only a small portion of variance ($\sim 5\text{--}6\%$, $R^2 \approx 0.053\text{--}0.058$)
 - Combined model improves explanation of variance ($R^2 = 0.164$, $RMSE = 0.70$)
 - Feed rate is the primary driver of increased variability in surface roughness, while other factors have secondary and interaction-dependent effects

12.4 Final Interpretation

Variance in surface roughness is primarily explained within the model by feed rate, with higher feed rates leading to increased variability in surface roughness.

13. Final Treatment of Non-Constant Variance in the Model

13.1 Rationale for the Final Approach

Initial diagnostic checks (residual plots and variance regression analyses) showed clear evidence of heteroscedasticity:

- Residual variance increases with feed rate
- Cutting speed ($vc = 390$) also shows elevated variability
- Depth of cut has a weaker and less consistent effect

A standard homoscedastic mixed model would therefore underestimate variability at higher factor levels and potentially inflate Type I error rates.

13.2 Final Variance Structure

To account for non-constant variance, a heteroscedastic mixed-effects model is specified using a variance function that allows different residual variances across feed rate levels:

$$\text{Var}(\varepsilon_{ijkl}) = \sigma^2 \cdot f^{2\delta}$$

This structure allows the error variance to differ depending on the level of feed rate, capturing the dominant source of heteroscedasticity identified in the data.

13.3 Final Model Specification

The final model is a linear mixed-effects model used to explain surface roughness (Ra) while accounting for both mean structure and changing variability.

- Fixed effects: ap, vc, f and selected interactions
- Random effects: Run (to capture run-to-run variability)
- Heteroskedastic variance structure: variance increases with feed rate

$$y_{ijkl} = \mu + A_i + B_j + C_k + (AB)_{ij} + (AC)_{ik} + (BC)_{jk} + b_l + \varepsilon_{ijkl}$$

where:

$$b_l \sim N(0, \sigma_{Run}^2)$$

and the residual variance is modeled as a function of feed rate rather than as separate variance estimates for each level:

$$\text{Var}(\varepsilon_{ijkl}) = \sigma^2 \cdot f^{2\delta}$$

13.4 R Implementation

```
In [89]: library(nlme)

data$f_num <- as.numeric(as.character(data$f)) # create numeric feed rate

final_model <- lme(
  Ra ~ ap * vc * f,           # mean structure (fixed effects)
  random = ~1 | Run,         # random run-to-run variation
  weights = varPower(form = ~ f_num), # variance increases with feed rate
  data = data,
  method = "REML"
)

anova(final_model)
```

A anova.lme: 8 × 4

	numDF	denDF	F-value	p-value
	<int>	<dbl>	<dbl>	<dbl>
(Intercept)	1	270	780.6160607	0.000000e+00
ap	2	27	6.6672620	4.433760e-03
vc	2	27	1.8498516	1.766442e-01
f	2	27	47.8733063	1.323678e-09
ap:vc	4	27	2.0925341	1.095255e-01
ap:f	4	27	1.2909527	2.982983e-01
vc:f	4	27	0.3756488	8.239421e-01
ap:vc:f	8	27	0.6447941	7.335184e-01

The code above presents the R implementation of the final mixed-effects model used in this analysis.

```
In [ ]: library(emmeans)

ap_emm <- emmeans(final_model, ~ ap)
vc_emm <- emmeans(final_model, ~ vc)
f_emm  <- emmeans(final_model, ~ f)

ap_df <- as.data.frame(emmeans(final_model, ~ ap))
vc_df <- as.data.frame(emmeans(final_model, ~ vc))
f_df  <- as.data.frame(emmeans(final_model, ~ f))
```

```
In [93]: ap_df
vc_df
f_df
```

A summary_emm: 3 × 6

	ap	emmean	SE	df	lower.CL	upper.CL
	<fct>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	0.25	0.8108889	0.04412397	27	0.7203540	0.9014238
2	0.5	0.5880370	0.04412397	27	0.4975021	0.6785719
3	0.8	0.7478426	0.04412397	27	0.6573077	0.8383775

A summary_emm: 3 × 6

	vc	emmean	SE	df	lower.CL	upper.CL
	<fct>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	310	0.7306389	0.04412397	27	0.6401040	0.8211738
2	350	0.7671852	0.04412397	27	0.6766503	0.8577201
3	390	0.6489444	0.04412397	27	0.5584095	0.7394794

A summary_emm: 3 × 6

	f	emmean	SE	df	lower.CL	upper.CL
	<fct>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	0.07	0.4105093	0.04374951	27	0.3207427	0.5002758
2	0.1	0.7144352	0.04401660	27	0.6241206	0.8047498
3	0.13	1.0218241	0.04460149	27	0.9303094	1.1133388

We also compute the estimated marginal means of surface roughness (Ra) for each level of the machining parameters. These values represent the model-adjusted average Ra while accounting for the effects of other variables, interactions, and run-to-run variability. This allows direct comparison of how ap, vc, and f individually influence the mean surface finish.

13.6 Interpretation of Final Model

Feed rate is the most statistically significant factor influencing mean surface roughness:

$$F_f = 47.873, \quad p = 1.32 \times 10^{-9}$$

followed by depth of cut:

$$F_{ap} = 6.667, \quad p = 4.43 \times 10^{-3}$$

while cutting speed is not statistically significant:

$$F_{vc} = 1.850, \quad p = 0.177$$

Interaction effects between machining parameters are not statistically significant at the 95% confidence level:

- $ap \times vc$: $p = 0.110$
- $ap \times f$: $p = 0.298$
- $vc \times f$: $p = 0.824$
- $ap \times vc \times f$: $p = 0.734$

Estimated marginal means further confirm the effect of feed rate on mean surface roughness:

$$Ra_{0.07} = 0.411, \quad Ra_{0.10} = 0.714, \quad Ra_{0.13} = 1.022$$

Depth of cut shows a non-linear effect:

$$Ra_{0.25} = 0.811, \quad Ra_{0.5} = 0.588, \quad Ra_{0.8} = 0.748$$

Cutting speed has a weaker and non-monotonic influence:

$$Ra_{310} = 0.731, \quad Ra_{350} = 0.767, \quad Ra_{390} = 0.649$$

Overall, feed rate is the dominant factor controlling mean surface roughness, depth of cut has a secondary effect, and cutting speed along with interaction terms contribute minimally to the mean response.

14. Final Summary and Concluding Remarks

This study analyzed the relationship between machining parameters—depth of cut (a_p), cutting speed (v_c), and feed rate (f)—and surface roughness (R_a) using a mixed-effects factorial modeling framework with an explicit treatment of heteroscedasticity.

14.1 Summary of Main Findings

The results consistently show a clear structure in how machining parameters affect surface roughness:

- **Feed rate (f)** is the dominant factor, significantly influencing both the mean and variability of surface roughness ($p < 0.001$).
- **Depth of cut (a_p)** has a statistically significant but weaker effect on the mean response.
- **Cutting speed (v_c)** does not have a statistically significant effect within the studied range.

No significant interaction effects were detected, suggesting that the factors operate largely independently in their influence on surface roughness.

14.2 Variance Structure and Model Adequacy

Diagnostic analysis revealed that the assumption of constant variance was violated, with variability increasing across machining conditions—particularly with higher feed rates.

This was addressed using a heteroscedastic mixed-effects model, allowing residual variance to vary by feed rate. This adjustment:

- Improved model fit (lower AIC)
- Produced more reliable standard errors
- Reduced bias in statistical inference

14.3 Key Statistical Insight

Accounting for non-constant variance did not change the overall conclusions about factor importance, but it strengthened the reliability of the inference.

Feed rate remains the primary driver of both the level and variability of surface roughness, while other factors play secondary or negligible roles.

14.4 Final Conclusion

The machining process is best described by a model in which feed rate governs both the expected value and variability of surface roughness. Depth of cut contributes moderately to the mean response, while cutting speed has minimal influence.

The final mixed-effects model with a structured variance component provides a robust and interpretable representation of the system, capturing both mean behavior and heteroscedasticity in a statistically coherent framework.

Overall, this analysis demonstrates the importance of accounting for variance heterogeneity in factorial experiments. Ignoring such structure would lead to less reliable inference, particularly in systems where process parameters directly influence variability, as observed here with feed rate.